

Week Summary

Linear algebra and matrices

Summary

- Many numerical methods used in HPC require solving systems of linear equations
- The performance is affected by whether the matrices are dense or sparse
Dense vs. sparse matrices:
 - Dense: BLAS and arithmetic intensity (high for BLAS level 3)
 - Sparse: compressed sparse row format (save space and computation, indirect addressing)
- Benchmarks: HPL and HPCG